STRUCTURAL AND PALEOMAGNETIC EVIDENCE FOR TERTIARY BENDING OF THE EASTERN BROOKS RANGE FLEXURE, ALASKA

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Abstract. Paleozoic carbonate sedimentary rocks from northeastern Alaska have been sampled for a structural and paleomagnetic study of the relationship of the Porcupine Terrane to both cratonic North America and adjacent Brooks Range terranes. Major structural features investigated includes the Yukon Porcupine megashear and the Northeast Alaskan oroclinal flexure.

Structural analysis of two well-exposed Paleozoic sections along the upper Porcupine River in northeastern Alaska yields estimates of shortening directions for two distinct phases of deformation. In the Upper Ramparts region of the Porcupine Terrane, north-south shortening (D1) occurs in Silurian-Devonian units. In the equivalent units of the Lower Ramparts northwest-southeast shortening occurs.

A much younger deformational event (D2), presumably associated with the Late Cretaceous Brookian Orogeny, overprints these earlier pre-Permian (D1) structural elements. In the Upper Ramparts the orientation of this event is somewhat equivocal but appears to be oriented northwest-southeast with vergence to the southeast. The Lower Rampart Permian section is clearly deformed by the D2 event in an east-west sense with vergence to the west. The general trend between the D1 and D2 deformational events differ by 45 degrees in the Upper Rampart Region. A similar variance between D1 and D2 is observed in the Lower Ramparts, but the structural orientations differ by 45 degrees between the Upper and Lower Ramparts, suggesting large scale block rotations occurred after the D2 shortening event.

The characteristic remanent magnetization postdates the latest phase of deformation (D2) and therefore represents an overprint. This overprint is presumed to be early to middle Tertiary, although the mean paleomagnetic pole lies off the expected apparent polar wander path (APWP) for North America. This paleomagnetic discordance is in declination only, and the sense and magnitude of rotation is clockwise approximately 45 to 90 degrees. It is proposed that the structural-grain of the Brookian Orogeny in northeastern Alaska was aligned with the northwest Canadian Cordilleran. Subsequent bending in the Tertiary produced the eastern Brooks Range orocline.

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Introduction

One of the most remarkable features of northern Alaska geology is the relatively sharp flexure of the fold and thrust belt that forms the northeast Brooks Range (Figure 1). This belt trends in an east-west direction as far east as the Romanzof Mountains (Figure 2). There it turns and trends in a north-south direction following the lineament formed by the Davidson and Ogilvie Mountains and the Kandik Thrust Belt. This sharp flexure in structural grain of the Brooks Range is continuously traceable around the northern and eastern edges of the Yukon Flats basin and the Porcupine Terrane.

The kinematic evolution of the northeastern Brooks Range flexure has been described by Patton and Tailleur [1977] as an effect of convergence between the rigid crustal plates of North American and Eurasia. Other regional mechanisms, which may be partly responsible for the observations of deflexed structural trends in the Northeastern Brooks Range are dextral shearing along the Yukon-Porcupine lineament (Figure 2) and oroclinal bending as a product of A-type (or Alpine-type) subduction [Bally, 1975].

Oroclines of similar scale to the northeastern Brooks Range flexure have been attributed to movement of wrench faults (see Marshak, [1988], for an excellent review). The Yukon-Porcupine lineament, which in northeast Alaska represents the transition zone between the deformed Paleozoic shelf carbonates of the northern Canadian Cordilleran and the accreted terranes of northern Alaska [Jones et al., 1986], bisects the southeastern limb of the orocline (Figure 1). In this region the structural fabric of the

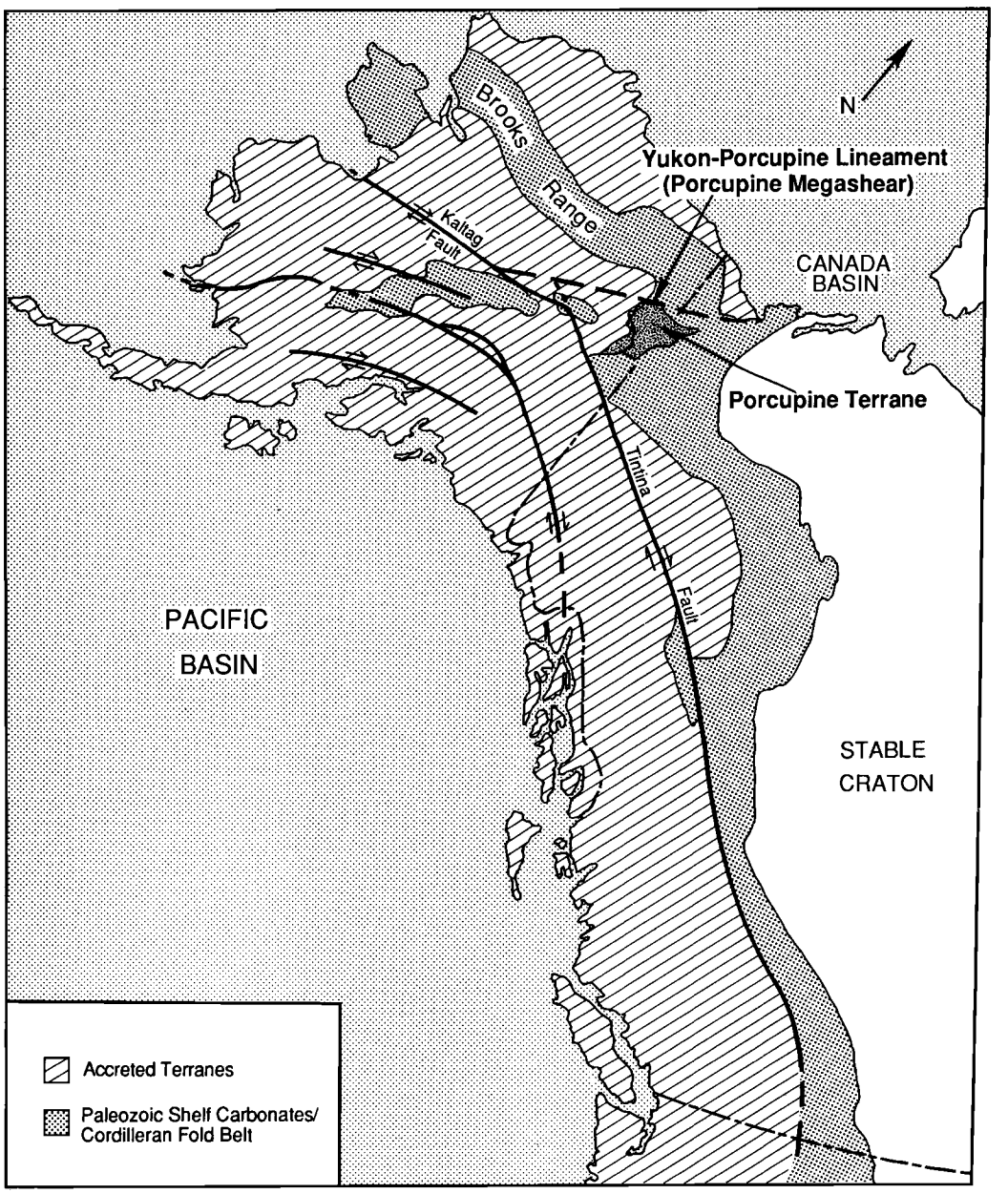


Fig. 1. General Paleozoic trends of Northwest North America and the location of the Porcupine Terrane study area.

Cordillera is bent in an apparent clockwise sense with respect to structural trends of the central Brooks Range and the northern Canadian Cordillera. The lineament, which is aligned with the right-lateral Kaltag fault zone of central Alaska (Figure 1) could be a wrench fault and therefore responsible for affecting the orientation of structural fabric if sufficient offset in a right-lateral sense is discovered. The tight oroclinal bend of the structural trends of the Brookian Orogeny in northeastern Alaska could be viewed as a massive drag fold, or alternately, as a wide region of distributed shear. Zones of distributed right-lateral shear segment crustal blocks into roller bearings and produce clockwise rotations [Beck, 1976, 1980].

Alternatively, the orocline could be caused by the outward (northward) migration and concomitant flexural bending of the deformational front as a consequence of A-type subduction. This involves the underthrusting of continental crust below continental crust and is often accompanied by an arcuate belt of compression and an inner region of extension and subsidence. This mechanism,

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comparable to those documented in other orogenic belts such as the Carpathians [Burchfiel and Royden, 1982] and Apennines [Malinverno and Ryan, 1986; Royden et al., 1987], would be accompanied by crustal stretching in the inner part of the eastern Brooks Range.

Our work addresses the origin of the northeastern Brooks Range flexure. We will show that the flexure is not a feature of the Brookian Orogeny, but of a later event superimposed on the Brookian structural trends. Our work does not bear on the problem of the Jura-Cretaceous evolution of the Arctic Ocean [Halgedahl and Jarrard, 1987; Crane, 1987; McWhae, 1986] and the hypothesized opening of the Canada basin [Tailleur, 1973]; rather it focuses on the effect of Tertiary tectonics on the northwestern Cordillera of North America. A further objective of this study was to determine from paleomagnetic poles whether or not the Porcupine Terrane was an allochthonous exotic block of Paleozoic carbonate or a deformed autochthonous shelf fragment of the northern Canadian cordillera.

The Porcupine Terrane is in northeastern (Figure 2) Alaska and extends a short distance into Canada [Jones et al., 1986, 1987]. This

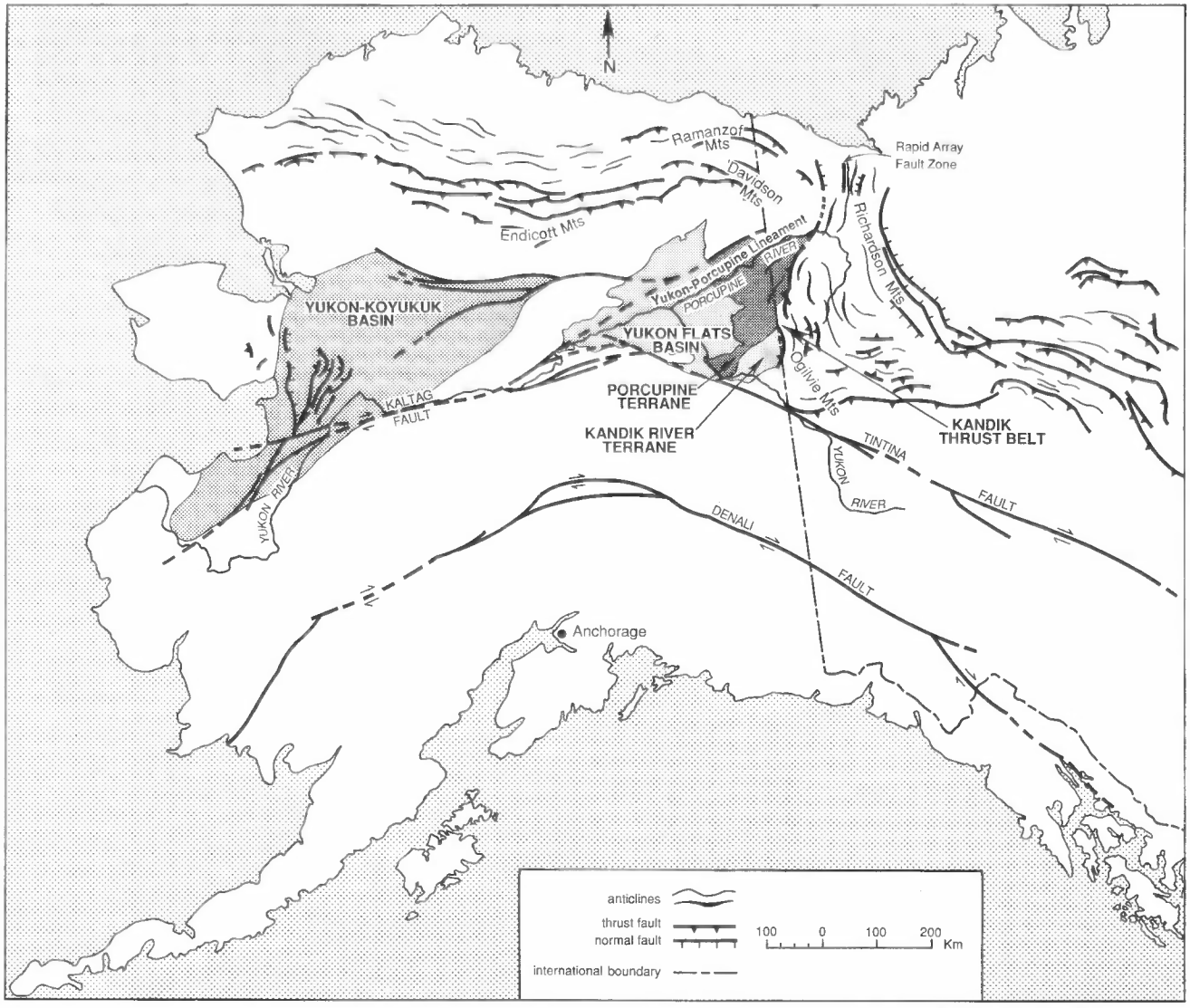


Fig. 2. Map showing the major contacts and faults in east-central Alaska around the Porcupine Terrane. Major interior low-lying sedimentary basins shown as light-shaded regions are separated by the Ruby geanticline. Dark-shaded region is the Porcupine Terrane. The Yukon Flats basin represents a Tertiary extensional region in the core of the eastern Brooks Range oroclinal flexure. Structural trends of the Eastern Brooks Range flexure are emphasized.

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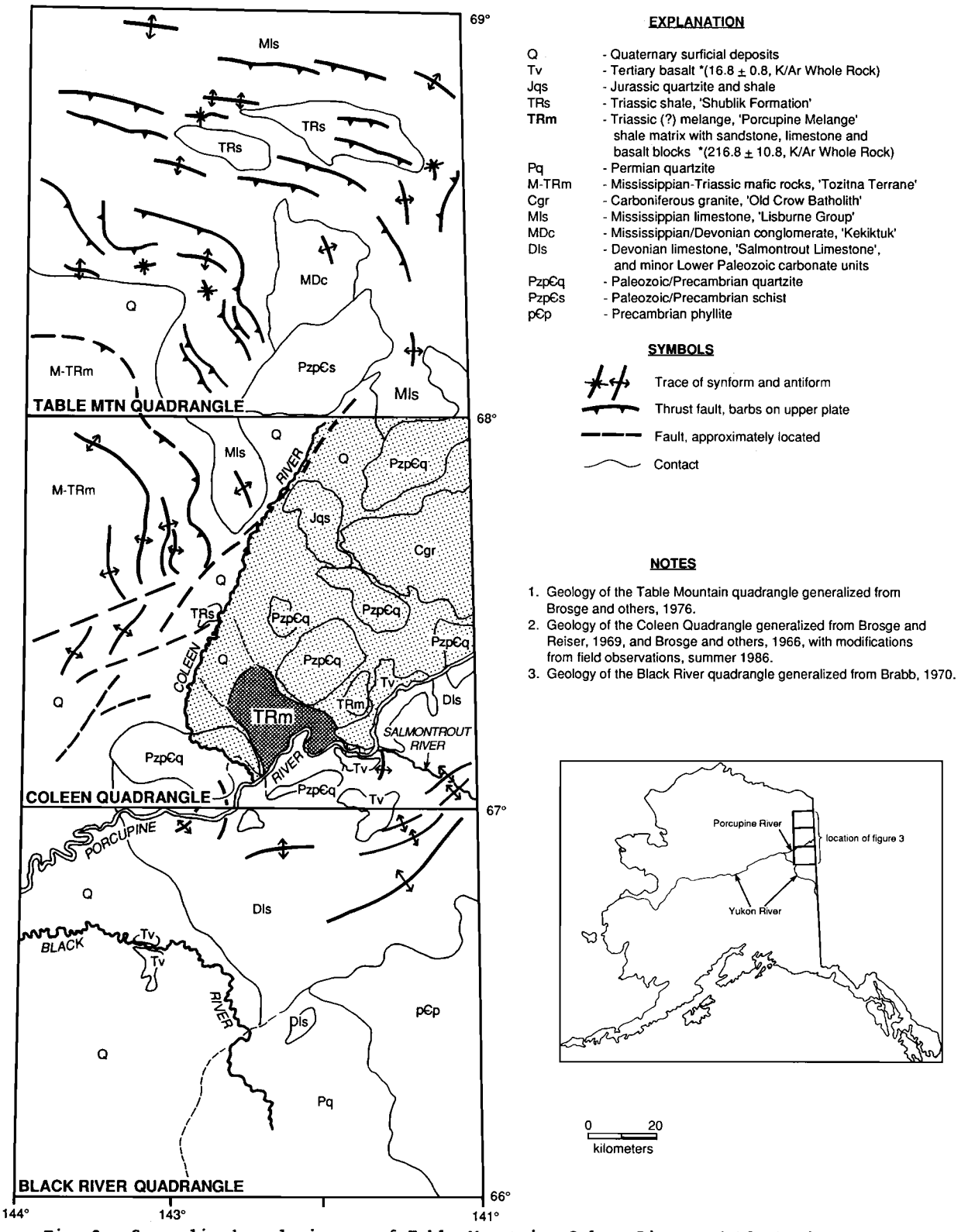


Fig. 3. Generalized geologic map of Table Mountain, Coleen River and Black River quadrangles. Dark-shaded region is the approximate location of the Porcupine Melange. Light-shaded region is a zone of distributed shear (Porcupine megashear) between the Porcupine Terrane and the neighboring Brooks Range terranes.

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trapezoidal terrane is bordered on two sides by major fault contacts and on the third by an overlap sequence [Howell and Wiley, 1987]. The northern boundary of the Porcupine Terrane trends northeasterly and generally follows the natural lineament defined by the Yukon and Porcupine Rivers. This lineament is thought to represent a major strike-slip fault, the Kaltag-Porcupine fault, otherwise known as the Porcupine megashear [Churkin et al., 1980]. Traversing in a northeast direction along this northern boundary, the Porcupine Terrane contacts the eastern edge of first the Tozitna Terrane, then the Endicott Mountains Terrane, and finally the North Slope Terrane. The eastern border of the Porcupine is in thrust contact (Kandik Thrust Belt) with the Paleozoic shelf sediments of the Canadian Cordillera [Norris, 1985; Monger and Berg, 1987]. The western border is buried under Tertiary cover of the Yukon Flats basin.

General Geology and Previous Work

The Porcupine River area was first studied by Kindle in 1908. In his reconnaissance of the Porcupine Valley, he delineated two large anticlinoria named the Upper and Lower Ramparts. Between these two uplifts lies the Coleen Basin: a low, nearly level, basin occupied by the Coleen River (Figure 3). Paleozoic strata is well-exposed along the Porcupine River canyon walls and its tributaries in the Upper and Lower Ramparts. Excellent exposures of Mesozoic rock units in the Coleen basin can be observed in the bed of the Porcupine River during low water intervals.

A major normal fault, with dip-slip displacement of not less than 1,000 meters, crosses the Porcupine River at the west entrance to the Lower Ramparts [kindle, 1908] and parallels the trend of the Lower Rampart anticlinorium. Here, Permian and Carboniferous shale structurally juxtaposed against Ordovician-Silurian limestone, dips to the west beneath the Tertiary cover of the Yukon Flats basin.

Carbonate rocks underlain by Silurian graptolitic black shales were first noted by Kindle at the confluence of the Salmontrout and Porcupine Rivers. The carbonate strata, formally named the Salmontrout Limestone, is massively jointed, light grey to blue limestone, which in turn is overlain by brownish colored shale. Most work since Kindle's has been concentrated on the biostratigraphy of the lower Paleozoic, especially the Salmontrout Limestone.

Work by Churkin and Brabb [1968] found the basal black shale to be Lower Devonian with the lowest bed being Silurian in age. This unit has been assigned to the Road River Formation, an Ordovician-Devonian siliceous carbonate and shale sequence, recognized in the Paleozoic terranes of east-central Alaska and the Yukon Territory.

The age of the Salmontrout was originally set as Middle Devonian by Kindle, but was redefined by both Churkin and Brabb [1968], and by Lane and Ormiston [1979], as Lower Devonian, although they disagree on the exact age. The overlying brownish colored shale has been reassigned to Lower Middle Devonian [Churkin and Brabb, 1968] and named the Old Camp Formation by Lane and Ormiston [1979].

Laudon et al. [1966], mapping the upper reaches of the Salmontrout River, reported approximately 5,000 feet of Upper Carboniferous and Permian strata that rest unconformably on the Devonian Salmontrout Limestone. They described this unconformity as "an essentially irregular solution-type topography that developed on the top of the Salmontrout Limestone". Notably, the Old Camp Formation is missing from their section.

Three other studies in the Porcupine area are the geologic reconnaissance map and stratigraphic sections produced by Brosge' et al. [1966], and the studies of a late Paleozoic orogeny in the adjacent northern Yukon [Bell, 1974, Norris, 1974]. Our current work relies greatly on age assignments reported in the former study.

Investigating the easternmost extension of the Porcupine Terrane into western Canada, Bell found a regional unconformity exposed beneath Carboniferous rocks. The units underlying this unconformity range from Precambrian to Upper Devonian and have been divided into three tectonic zones. The central zone correlates with the Porcupine Terrane and exhibits faulting and folding about northeast-trending axes. Bell interprets the late Paleozoic orogeny to have involved either or a combination of: a) southeasterly directed overthrusting, b) strike-slip faulting along a northeast-trending fracture zone and/or c) block faulting.

Structural Analysis

Observations of the structural style of deformation were made on outcrops of Paleozoic rocks along the Porcupine River in the Upper Ramparts and Lower Ramparts regions (Figure 4). In these areas, almost continuous sections of bedrock crop out along the river and present an excellent transect for studying geometrical relationships, stratigraphy, and regional unconformities. The structures recorded include bedding attitudes; fold axes; normal, reverse, and strike-slip faults; and foliations. Paleomagnetic sample sites, noted on Figure 4, coincide with regions in the Upper and Lower Ramparts where the structural style of the Paleozoic strata is relatively simple and both assigned age and paleohorizontal are known (Figure 5).

Structural Style, Upper Ramparts

An excellent exposure of Siluro-Devonian strata is present at the intersection of the Salmontrout and Porcupine Rivers. Tectonic deformation appears to have greatly increased the

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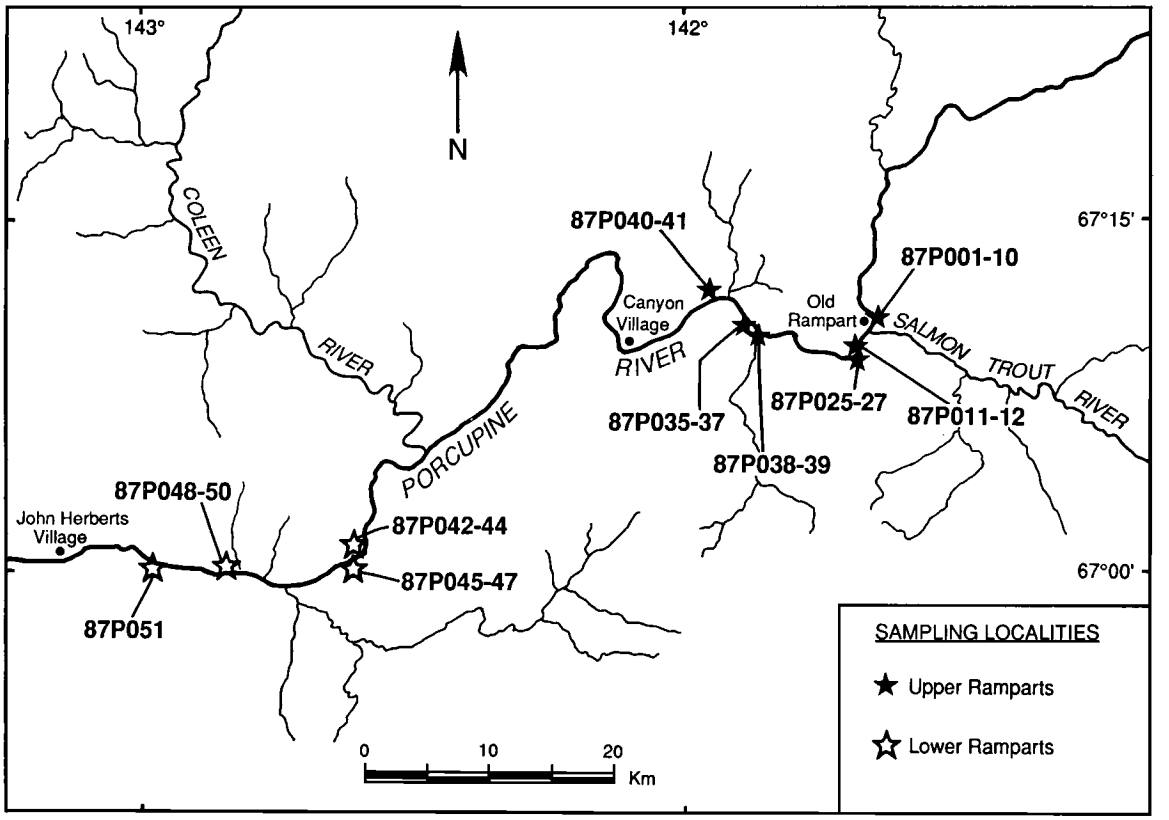


Fig. 4. Detailed location map of the upper Porcupine River area and paleomagnetic sampling localities for the Upper (Solid Stars) and Lower (Open Stars) Ramparts.

thickness of the exposed carbonate beds. A major thrust slices through the section producing a closely-spaced fracture cleavage at a high angle to bedding as well as subhorizontal brecciation zones of varying widths from meters to tens of meters. Many of the macrofossils are truncated by pressure solution along subvertical fractures. In this area, the thickness of any single carbonate bed is extremely variable and often uneven in appearance. This is believed to be a result of tectonic deformation. It is inferred that the maximum principal stress for this event was oriented north-south and was subhorizontal.

As a consequence of polyphase deformation, large asymmetric folds and monoclinal flexures in this region have both variable wavelengths and highly variable axial orientations. Outcrops of shale and siltstone of Mississippian age located opposite the mouth of the Salmontrout River offer clear evidence for interference patterns between two different sets of structures (Figure 6). Isoclinal folds (F1) are bent around the trace of the axial surface of F2 folds. The F2 fold system is clearly superimposed upon the isoclinal fold pattern, F1.

Numerous isoclinally folded Mississippian siltstone beds were observed along the western half of the Upper Ramparts river cuts. These are similar in orientation to the major structural elements seen in the underlying Devonian carbonate. However, a highly irregular surface with a 2 meter-thick zone of intense shearing separates the Mississippian strata from the Devonian strata. This contact has been described as an unconformity by Brosge' et al. [1966] and by Laudon et al. [1966], but it could alternately represent a structural boundary between folded units of brittle-deforming Devonian carbonate rocks and the overlying ductile shale.

Structural Style, Lower Ramparts

The style of deformation in the Lower Ramparts section of the Porcupine River does not differ substantially from that of the Upper Ramparts. Numerous large-scale reverse faults and associated minor structures were observed throughout this section. Strike-slip faults of indeterminate displacement also occur along the outcrops of the Lower Ramparts. Many of these strike northwest to north and have fault gouge zones up to 5 meters in width with subhorizontal striations. Slickensides and geomorphologic indicators suggest left-lateral displacement.

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Porcupine Megashear and the Porcupine Melange

The Porcupine megashear is a pie-shaped area of chaotically mixed and fragmented subterranes. This wide zone opens northeastward and is defined by the Coleen River on the north, by the Porcupine River on the south, and lies in the center of the Coleen River Quadrangle (Figure 3). It includes blocks of Precambrian quartzites, Paleozoic carbonate platform rocks, a large Late Paleozoic batholith, and a newly recognized melange unit--the Porcupine Melange. The fragmented subterranes of the Porcupine megashear have lithotectonic similarities to all major tectonostratigraphic terranes which border it (Figure 3), whereas the various components of the newly named Porcupine Melange, are only remotely similar to lithologies of the oceanic Tozitna Terrane that lies to the north.

The Porcupine Melange lies within the borders of the Upper and Lower Ramparts (Figure 5), and contains both coherent and noncoherent subunits. The coherent blocks of the melange had been previously recognized by Kindle [1908] and later by Brosge' et al. [1966], but taken together, were not considered a mappable unit. The Porcupine Melange is a structurally complex assemblage of unaltered lens-shaped basalt blocks tens of meters in length and minor carbonate and sandstone blocks, all of which are surrounded by a matrix of highly deformed and mildly metamorphosed shale. Shear bands of dark chloritic metamafics and brown-yellow felsic lenses of varying lithologies are commonly interfingered in a chaotic manner and display an extraordinary "tiger stripe" structure. The predominant mesoscopic structures developed in the shale are tight chevron folds with nearly vertical fold axes.

Distinctive paleontologic specimens have not been found in the sedimentary blocks; however, a whole rock K/Ar date of 216 ± 10.8 my. (Late Triassic) was determined from a massive basalt flow in a coherent section located in the northeast portion of the melange (Figure 5). The geographic location of the dated flow is at the entrance to Howling Dog Canyon in the Upper Ramparts region of the Porcupine River (141°58'W, 67°11'N).

Structural Orientation Data

Attitudes of structural features were collected at regular intervals along the study transect and at all minor isolated rock exposures. These directional data, when averaged and contoured, help evaluate the orientation of paleo-principal stresses, shortening directions, and phases of deformation. Although the deformation of the continental margin in the northern Cordillera is quite diffuse, estimates of convergence derived from the integration of strain markers should provide information on the nature of the boundary conditions which existed between terranes in northern Alaska.

Most fabric elements in pre-permian strata show a large dispersion in orientation due to the interference of two phases of deformation. In our analysis, attitudes of the first phase structures have not been corrected by removing the effects of the second phase of deformation. Additionally, pre-Permian data sets may contain attitudes (ie. fold axes) that are from two separate generations of deformation.

The orientation data of the structural elements developed in Silurian and Devonian rocks are plotted separately from those of the

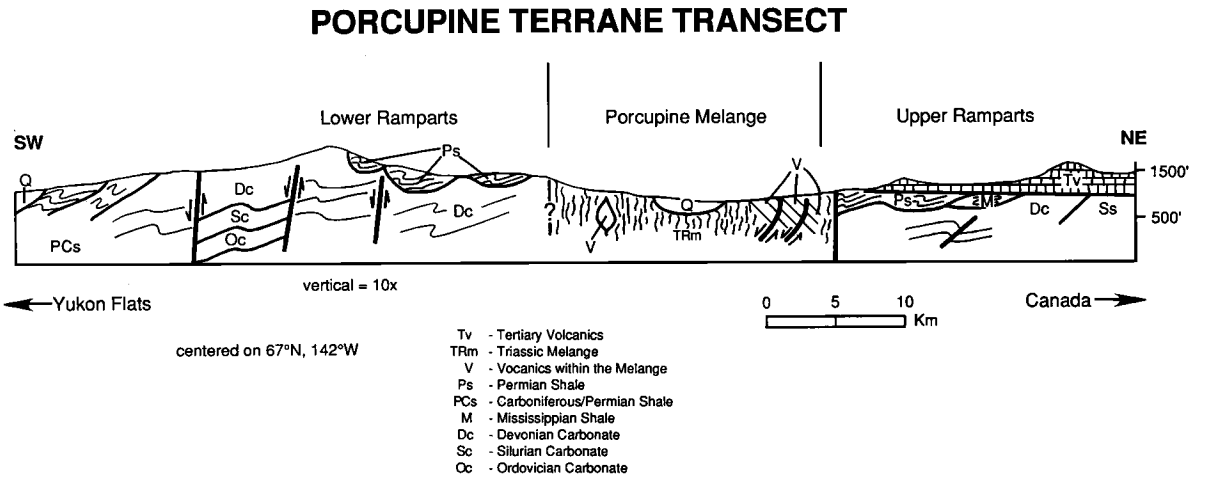


Fig. 5. Schematic cross section of structural traverse of the upper Porcupine River from Lower Ramparts on the left to Upper Ramparts on the right.

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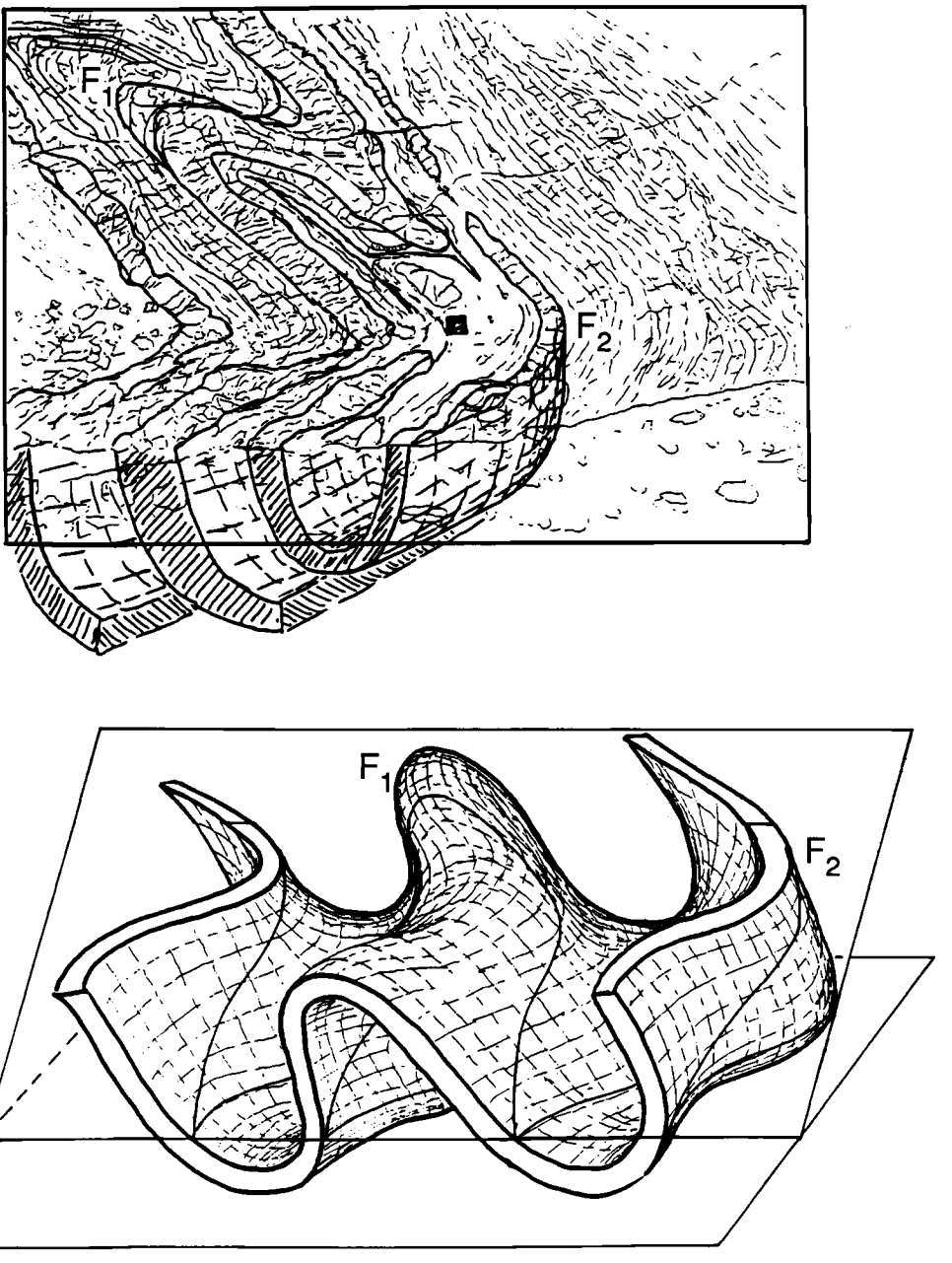


Fig. 6. Sketch of Mississippian polydeformed section showing location of F1 and F2 folds (top), and interpretive diagram (bottom). Planes on lower figure represent face of inclined outcrop and the horizontal riverbed. Width of sketched outcrop is approximately 5 meters.

Carboniferous and the Permian rocks respectively. This age separation facilitates the recognition and assignment of structural elements to their correct periods.

Upper Ramparts. From the attitudes of major thrusts and associated minor structures in the Siluro-Devonian section, we can deduce a north-south sense of lateral compression due to convergence. Supporting evidence for this sense of shortening is seen in the poles to bedding, S_o, which form a north-south girdle pattern (Figure 7). It is not clear whether this

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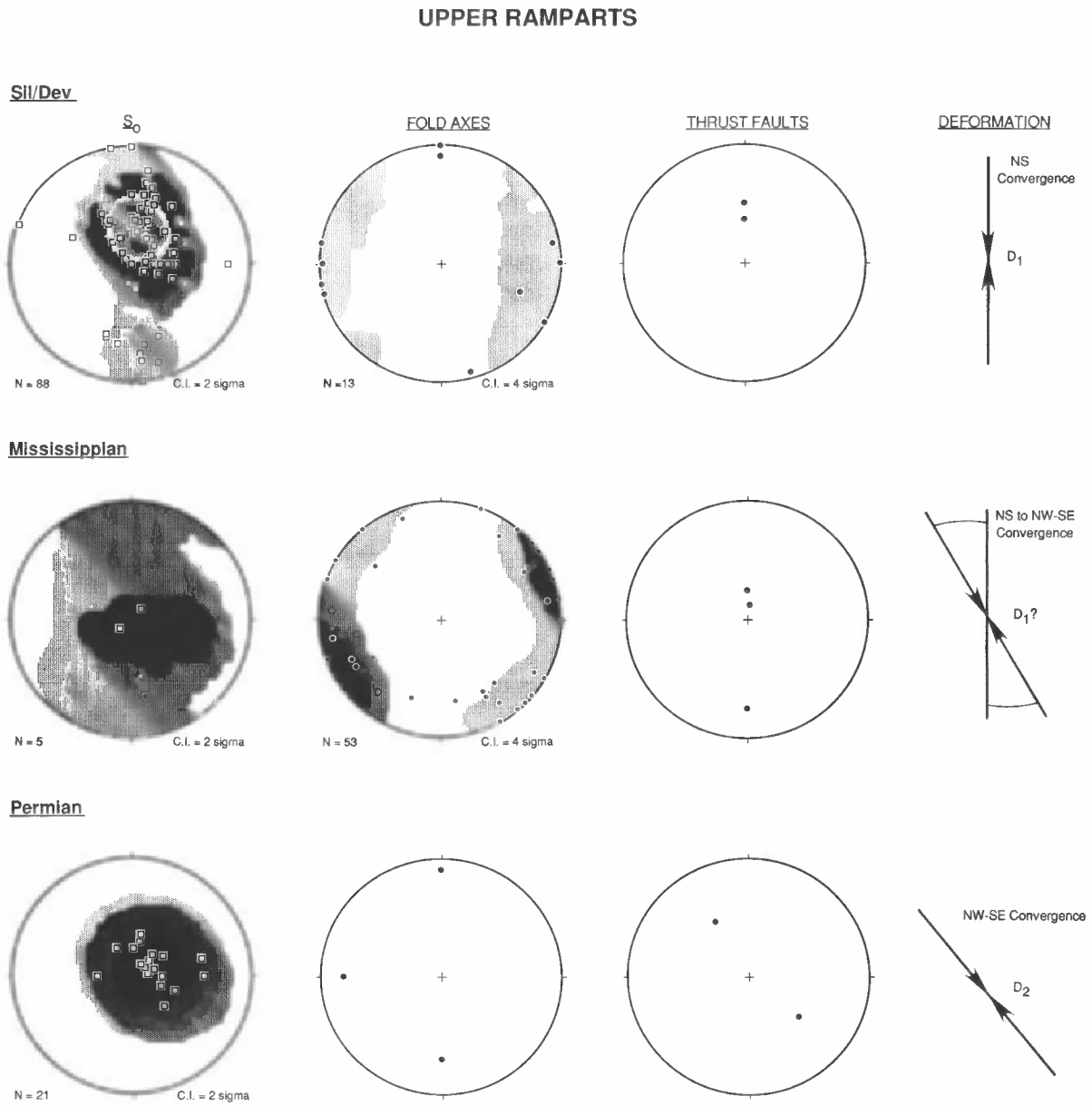


Fig. 7. Structural Analysis. Equal area projections of directional data for both Upper and Lower Ramparts. S_0 represents poles to bedding. Convergent arrows on right of plots indicate implied structural shortening direction. Observed densities are contoured at intervals of either 2 or 4 sigma using the method of Kamb [1959].

deformational event, D1, involved the Mississippian strata or not. From evidence of thrust faults alone we might conclude that the age of D1 deformation was post-Mississippian/Late Carboniferous, similar to the age of deformation seen to the south in the Nation Arch region [Payne and Allison, 1981]. But it is evident that the Mississippian is multiply deformed and was shortened more then the underlying Devonian carbonate sequence. Hence, if D1 is responsible for the pre-Carboniferous unconformity as noted by Brosge' et al. [1966] and Laudon et al. [1966], then an additional strong pre-Permian event, D1', is required.

The second-phase folds in the Mississippian strata relate to a D2 event. These folds are steeply inclined and disperse the first-phase east-west fold axes formed by D1 convergence into the northeast, southeast, southwest, and northwest quadrant clusters seen in the stereonet

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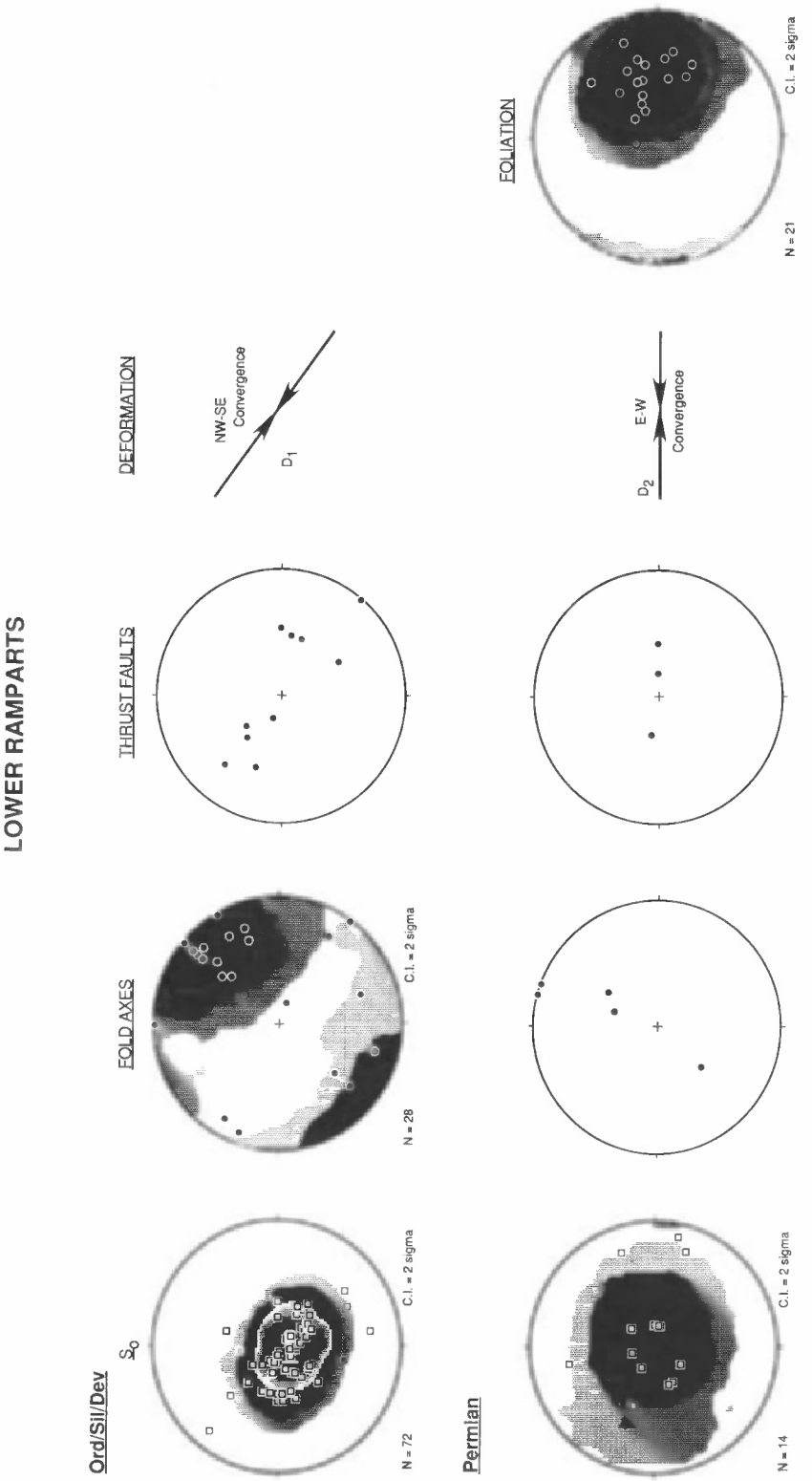


Fig. 7. Continued

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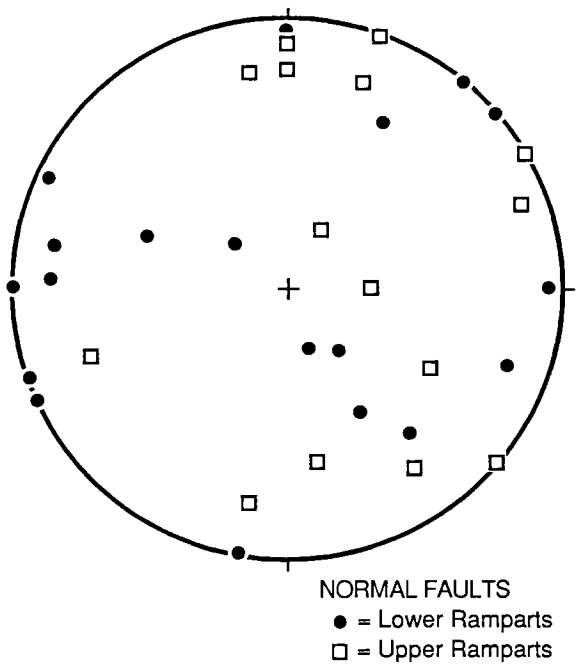


Fig. 8. Equal area plot of poles for normal faults of Upper Ramparts (open box) and Lower Ramparts (filled circle).

plot (Mississippian-Fold Axes) Figure 7. The axis attitudes of the second-phase folds in the Mississippian strata are not plotted.

Outcrops of Permian rocks in the Upper Ramparts are very limited. From thrust fault attitudes and bedding plane (S_0) plots we interpret a northwest-southeast sense of lateral compression. This event, D2, is post-Permian in age.

Lower Ramparts. The first phase deformation, D1, is very well-defined in the Devonian sections. Bedding (S_0), fold axes, and thrust fault attitudes support a northwest-southeast direction of lateral compression. Structural data from Mississippian strata were not collected in the Lower Ramparts. As in the Upper Ramparts, Permian rocks occur in only limited exposures; however, the degree of deformation is more intense and the shale exhibits an excellent axial planar foliation. The sense of shortening is east-west with vergence to the east (Figure 7).

The trends of D1 seen in the Lower Ramparts are similar to those seen in the Black River Quadrangle to the south and to trends in the pre-Permian fold belt mapped at Rats Pass in the Yukon territory [Bell, 1974]. The orientation of structural fabric of both D1 and D2 deformational events in the Upper Ramparts varies by 45 degrees in a clockwise sense to those documented in the Lower Ramparts (Figure 7).

While it is not uncommon for foldbelt trends to vary along strike, we believe it unlikely that the variation in trends of two separate events would mimic each other in two separate areas. The structural data from the Upper and Lower Ramparts suggest that both areas were equally affected by the D1 and D2 deformations, but that subsequent events led to a relative rotation between the two areas. This rotation might have been caused by the generation of the northeast Brookian flexure or by block rotation due to right-lateral shear in the Porcupine Megashear.

Normal Faults. Numerous normal faults cut all previously described structures. The majority of the faults dip toward the Yukon Flats basin to the west of the study area (Figure 8). The fault with the greatest amount of down-drop recognized by Kindle [1908] is at the west end of the Lower Ramparts section (Figure 5). Normal faults mapped in this study have minor amounts of strike-slip displacement while dip-slip displacement ranges from meters to tens of meters. We found no obvious difference comparing fault-plane attitudes between Upper and Lower Ramparts regions. The east-facing fault dips plotted on Figure 8 are believed to be antithetic to those that face west.

These normal faults are a result of crustal extension west of the Porcupine Terrane. We conclude from these data and structural data gleaned from neighboring quadrangles, that the Yukon Flats region (Figure 2) represents an extensional, rapidly deepening Tertiary basin. It is not clear if the extensional event inferred from these faults coincides with, or postdates, the apparent clockwise rotation of the Upper Ramparts relative to the Lower Ramparts. However, coinciding with this extensional event are volcanics with the geochemical affinity of continental flood basalts [Karl Wirth, Personal Communication, 1988]. A middle flow from one of the volcanic fields which covers the Upper Ramparts region yields a K-Ar date of 16.8 ± 0.8 ma (Figures 3 and 5). Preliminary paleomagnetic results from 17 flat-lying basalt flows indicate no evidence of tectonic rotation [Plumley and Vance, 1988]. Thus, the relative rotation between the Upper and Lower Ramparts must have occurred previous to the eruption of the volcanic fields.

Structural Summary

Two strong deformational events are recognized from regional relations seen in the Upper and Lower Ramparts region of the Porcupine River. We conclude from the structural style and vergence that the post-Permian D2 event is the late Mesozoic-Cenozoic Brookian Orogeny. The earlier D1 event may be post-Mississippian. We currently view the pie-shaped region contained by the Coleen and Porcupine Rivers (Figure 3) as a zone of distributed shear (Figure 9). This feature, previously named the Porcupine Megashear [Churkin et al. 1980] consists of numerous structural domains of Paleozoic strata which preserve the D1 and D2 deformational events. The various

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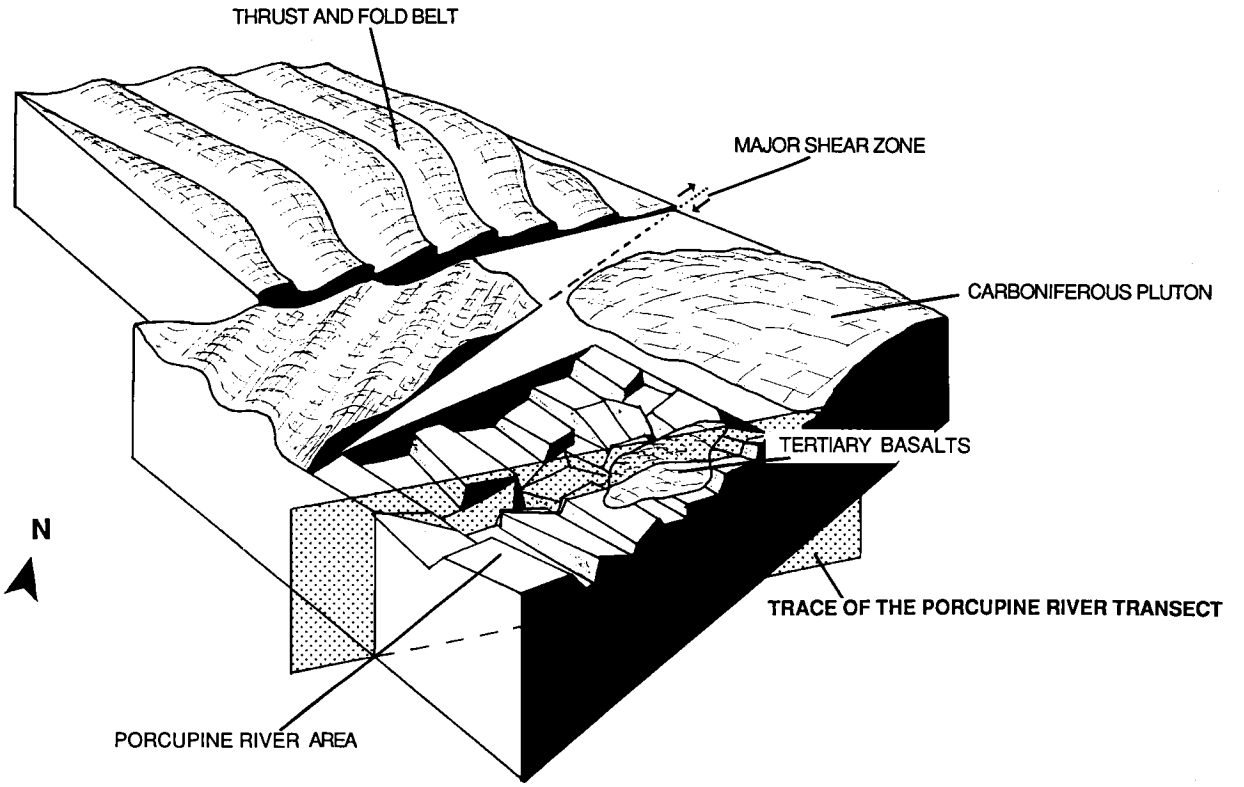


Fig. 9. Perspective block diagram of Porcupine River region showing the megashear zone, the Carboniferous pluton, overlying Tertiary basalts, and the trace of the structural transect shown in figure 5. Length of transect is 100 kilometers.

domains, of which two are the Upper and Lower Ramparts sections described above, apparently underwent relative rotations about vertical axes and then experienced an extensional event centered in the Yukon Flats basin as evidenced by the normal faults.

With the exception of the east-dipping normal faults, the orientation of the structural elements observed in the Porcupine Melange (a unique structural domain of the Porcupine megashear) differs substantially from that of the Paleozoic rocks cropping out in both the Upper and Lower Ramparts. Consequently we believe that the Triassic Porcupine Melange represents a displaced allochthonous block within the Porcupine Terrane.

Paleomagnetic Analysis

During the 1986 summer field season we sampled Silurian-Devonian carbonates, Carboniferous carbonates, Permian carbonates, Triassic basalt flows, and late Tertiary basalt flows for paleomagnetic study. The Silurian Road River Formation and the Devonian Salmontrout limestone (30 sites - 150 cores) and Tertiary basalt (19 sites - 110 cores) represent the majority of the collection. Of these studies, field tests (i.e. fold test, conglomerate test) necessary for determining the primacy of the remanence are available for the carbonates only. In this paper we report on the carbonate rock samples of Paleozoic age only.

Twenty sites were sampled in the Upper Ramparts section of the Porcupine Terrane and 10 from the Lower Ramparts (Figure 4). Samples from all 30 sites have been analyzed for paleomagnetic components. Demagnetization experiments include stepwise alternating field (AF) and stepwise thermal heating.

Paleomagnetic samples were drilled in the field with a gas-powered drill using non-magnetic stainless steel bits. Orientation of core azimuth was determined with a magnetic compass, and the regional magnetic declination was checked at all sites by back-sighting to major recognizable land forms. Our sampling strategy was to sample about six cores per site and one site per ten meters of stratigraphic section. The field cores were sectioned into samples with a nonmagnetic twin-blade saw, after which they were stored in a field-free environment for one month prior to measurement and demagnetization experiments.

Thermal demagnetization was attempted on pilot samples from every site. Samples were heated at

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intervals of 50 degrees in the range of 100°C to 300°C and then at 25 degree intervals up to the noise limit of our modified Schonstedt Spinner Magnetometer (SSM-2). The noise limit is $1x10^{-8}$ emu for a 30 minute measurement. Alternating field (AF) demagnetization was accomplished with a Sapphire Instruments model #SI-4, which is capable of handling peak fields of 1800 oersteds.

All samples underwent either AF or thermal stepwise demagnetization. A few were thermally demagnetized to 300°C followed by AF demagnetization to 1200 oersteds. Principal component analysis [Kirschvink, 1980] was used to compute the magnetic component for linear segments observed on orthogonal vector diagrams.

Paleomagnetic Results

Cognizant of the ubiquitous and stable Cretaceous overprints observed in the northern Brooks Range [Hillhouse and Gromme, 1983], we expected remagnetization in the Porcupine Terrane. Stable overprinting components from the Paleozoic D1 and Late Cretaceous D2 events were anticipated. An overprint related to a D1 event predating the Brookian Orogeny would be extremely useful for constraining the motion of Paleozoic carbonate terranes in Mesozoic times. At best, we hoped to recover primary components of magnetization and answer the questions regarding the latitudinal origin of the Porcupine Terrane.

To determine which demagnetization technique best isolates the characteristic component, an approximately equal number of samples were examined by progressive thermal demagnetization treatments (n=6) and AF demagnetization (n=7) from a representative site (86P044). While the components differ in direction (magnetic declination and inclination) for each sample and do not show any overlap in ranges of either blocking temperature or coercive force, the site-mean direction of the secondary overprint (Decl=332, Incl=83, a95=9.7) is similar to the characteristic component (Decl=291, Incl=86, a95=3.5). Likewise, the characteristic component site-mean direction derived from thermal treatment (Decl=297, Incl=85, a95=6.1) is statistically identical to that derived by AF treatment alone (Decl=284, Incl=86, a95=5.3). As AF treatment is less likely to affect the magnetic mineralogic content of the samples a majority of the sites were cleaned with stepwise AF.

Secondary and characteristic components of magnetization were measurable in most samples. The secondary components are usually removed in the temperature range of 100°C - 300°C whereas the stable characteristic component has a blocking temperature range of 300°C - 500°C. During AF treatment, the secondary component is normally removed after 200 oersted; the characteristic component may be present up to 1400 oersteds. Declinations for both components, before correction for bedding tilt, are typically north to northeast and inclinations are steeply down. Figure 10 illustrates both demagnetization behaviors for the magnetic cleaning experiments (sample # 86P007A and 86P0279B). Both secondary and characteristic components, isolated by either demagnetization treatment, are overprints and do not represent primary Paleozoic components of magnetization.

Samples from one site (86P038) from the Upper Ramparts region contain three distinct components of magnetization. The component first removed during demagnetization is similar in direction and blocking spectra to the secondary component observed in all other sites. Similarly, the second component determined from site 38 is similar in direction to the typical characteristic component of the other sites. However, the component isolated at both the highest blocking temperatures and/or alternating fields has a direction of magnetization, prior to bedding tilt correction, that is southwest and down (Figure 10, sample # 86P0214A). The significance of this third component will be discussed separately following presentation of the other results.

Sites 1 through 10 traverse a broad asymmetric syncline that underlies a major south-verging thrust. These sites have very different bedding plane attitudes (Table 1) and represent an excellent sequence for the application of the classic fold test [Graham, 1949]. The dispersion parameter (k) for these sites alone varies from k=66 in uncorrected geographic coordinates to k=5 in corrected stratigraphic coordinates. This suggests that both secondary and characteristic components were acquired after folding. As indicated by the kappa versus incremental tectonic correction plot (Figure 11), the stable overprinting magnetization probably completely post-dates the Brookian stage of lateral shortening and related deformation.

The mean characteristic directions (Table 2 and 3) of the Upper and Lower Ramparts are less than 5 degrees apart in geographic coordinates, and are not significantly distinct at the 95% confidence level (Figure 12). All sites were combined (n=30) to yield a mean direction in geographic coordinates (Decl=37.6, Incl=79.8, a95=3.7) and pole (Long=272.3 E, Lat=76.6 N, a95=6.4) for these Paleozoic carbonate rocks (Table 4).

Overprint Virtual Geomagnetic Poles and Secular Variation

While these rocks are clearly remagnetized the tight constraints on regional structural style of the Ramparts region allow significant conclusions to be drawn from the position of the mean geographic pole computed for this study. However, valid questions may be raised concerning the use of mono-polarity overprints which are clearly post-folding in acquisition. These include: what is the age of remanence

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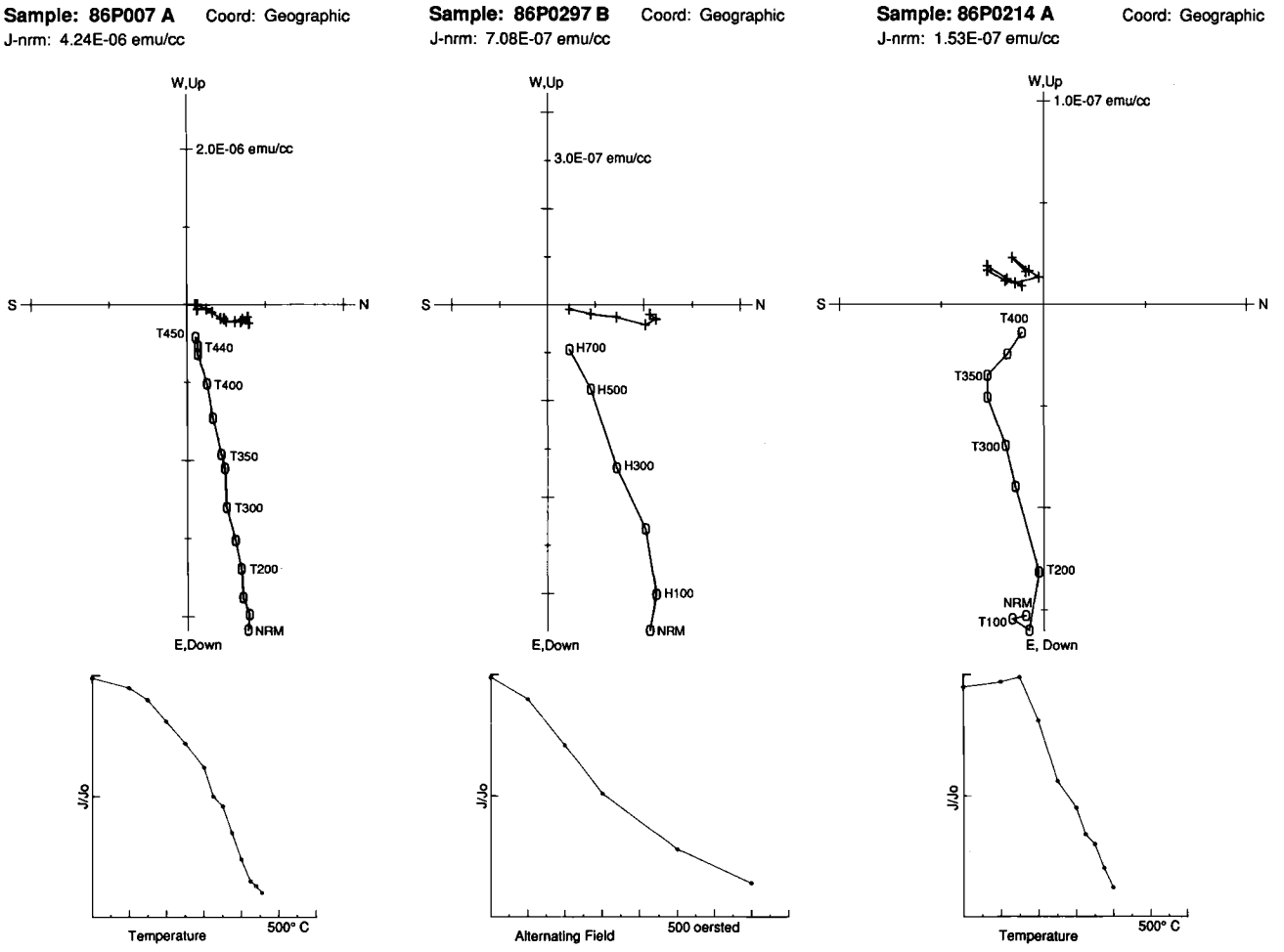


Fig. 10. Orthogonal vector-difference diagrams. Most carbonate samples have a simple two-component magnetization; both components are isolated by either alternating field (AF) or thermal experiments. Sample #86P007 illustrates stepwise thermal demagnetization, while sample #86P297B shows stepwise AF demagnetization. The three component magnetization illustrated in the demagnetization diagram of sample #86P0214A is representative of all samples at site 86P038. In addition to both previously determined components, these samples have a third stable component, isolated at moderately high blocking temperatures. Crosses represent the horizontal component, open circles the vertical component. J-nrm is the initial intensity of natural remanent magnetization. Units of magnetic intensity are in emu/cc.

acquisition? Have the strata undergone further undetected tilt since remanence acquisition? And will the average of the site-mean virtual geomagnetic poles (VGP) be a valid estimate of the geographic axis? While a discussion concerning the mechanism of remagnetization is also appropriate in response to these questions, we will concentrate our analysis to observations and interpretation of the angular dispersion of VGP.

The three dominant influences on the angular dispersion of overprint VGP in the carbonate rocks are the rate at which the characteristic component was blocked in, structural deformation other than homoclinal tilting following remanent acquisition, and the paleolatitude of the site. A recently proposed model [McFadden and McElhinny, 1984], which predicts paleosecular variation (angular dispersion of VGP) for the interval of 45-110 Ma, yields an estimate of 20.9° for the mean standard deviation for sites at latitudes greater than 60°. It may be concluded that if the angular dispersion of VGP lies greatly below this value that the sites have acquired their characteristic magnetizations over a period insufficient in length to average out paleosecular variation, whereas if the angular dispersion is significantly greater than the

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TABLE 1. Tilt Corrections for All Sites.

#	Site	Tectonic Corrections	#	Site	Tectonic Corrections
	Upper Ramparts		16	86P035	280/58
1	86P001	70/32	17	86P036	289/55
2	86P002	66/20 to 90/33	18	86P037	290/53
3	86P003	100/30	19	86P038	264/71
4	86P004	135/32 to 120/20	20	86P039	270/73
5	86P005	65/35		Lower Ramparts	
6	86P006	90/31	1	86P042	280/8
7	86P007	60/41 to 94/31	2	86P043	225/10 to 230/10
8	86P008	110/45	3	86P044	230/11
9	86P009	68/32 to 64/34	4	86P045	250/20
10	86P010	261/58 to 265/90	5	86P046	250/20
11	86P011	160/12 to 155/15	6	86P047	250/20
12	86P012	170/10	7	86P048	232/25 to 241/25
13	86P025	133/23	8	86P049	12/17 to 108/36
14	86P026	146/16 to 137/18	9	86P050	14/10 to 0/5
15	86P027	143/26	10	86P051	206/30

Note: Correction = Strike/Dip, where dip is always clockwise from strike.

prescribed value, then there is likely undetected tilt that has resulted in excess dispersion. The estimate of angular dispersion of geographic VGP for this study (Table 4) is 19.0° with upper and lower limits of 23.1 and 16.1 respectively [Cox, 1969]. The excellent agreement between the estimates of observed and predicted values of angular dispersion supports the validity of the resultant paleomagnetic pole and its further use in interpreting the kinematics of the Northeast Brooks Range flexure.

It is likely that true secular variation is recorded in the overprints; hence, timing of overprinting varies from site to site. This would suggest rapid blocking of the characteristic magnetization, probably because of the acquisition of a chemical remanent magnetization (CRM). In paleomagnetic studies of remagnetized carbonate sediments, the characteristic component is presumed to be either a chemical overprint [Scotese et al., 1982; McCabe et al., 1983, 1984] or a viscous-thermo remanent magnetization (VTRM) [Kent, 1985] and the secondary component is thought to be a partial-VTRM component that is blocked-in during the present normal polarity interval. In all cases, the high temperature/high coercive component in the remagnetized carbonate strata accurately records and retains the true direction of the ambient paleomagnetic field. The Paleozoic carbonate rocks of the Porcupine River are similar in that they show a two component magnetization, and in many samples, the characteristic component is still present after heating to 525°C. Such a scenario is suitable for this study; however, in this case, the secondary component was likely acquired shortly after the characteristic.

The combined geographic coordinate site-mean pole for the characteristic magnetization lies at 77.5° N, 276.5° E and does not overlie any segment of the expected Tertiary North American APWP (Figure 13). Both secondary and characteristic mean poles are significantly distinct from all computed Tertiary reference points (20 my, 30 my, and 60 my). While this pole position lies close to the Earth's present geomagnetic pole (78.5°N, 290°W), we do not attribute the observed stable remanence to the present day field, but instead, in consideration of the high blocking temperatures required to remove it, attribute it to a mid-Tertiary

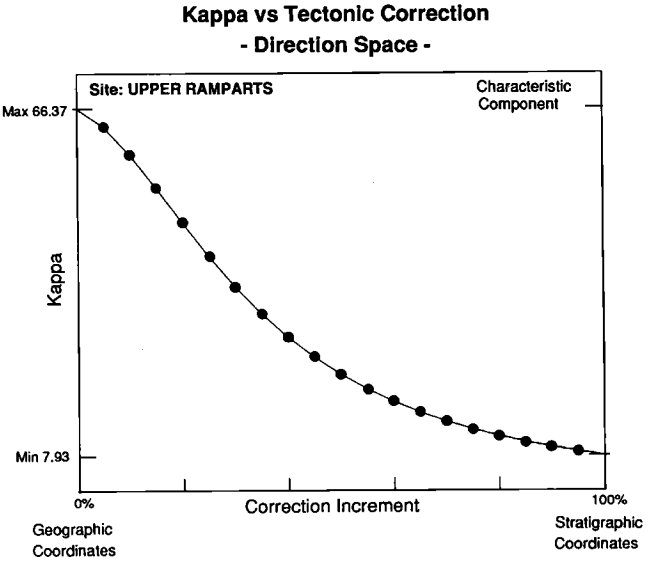


Fig. 11. Kappa versus incremental tectonic correction for Upper Ramparts, sites 86P001 through 86P010. Variation in dispersion of site-mean directions indicates acquisition of characteristic component is post-deformation in age.

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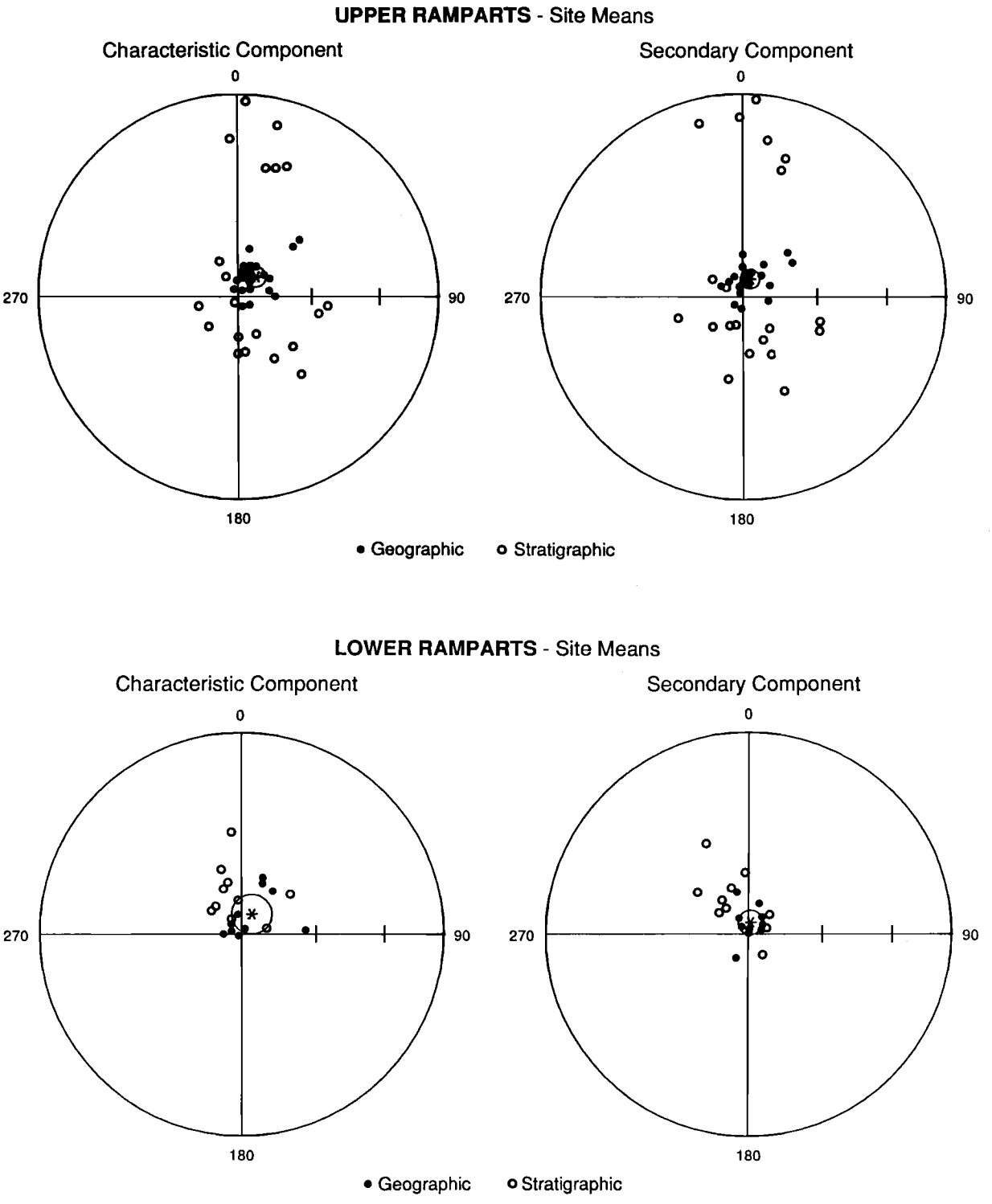


Fig. 12. Directional data for Upper Ramparts and Lower Ramparts paleomagnetic sites. Crosses represent geographic coordinates, filled circles represent stratigraphic coordinates. Star is the mean direction for geographic/stratigraphic plot with corresponding circle of confidence (a95). Projections are equal area.

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remagnetization event. There are many scenarios in which the computed mean geographic pole for this study could be displaced from the reference path of North America, however, it is helpful to first discuss the expected age of the characteristic and secondary components.

Most paleomagnetic overprints in the Brooks Range were acquired during or shortly after the Aptian in response to telescoping and shortening in the western and central part of the range [Mull, 1982; Mayfield et al., 1983; Hillhouse and Gromme, 1983; Plumley and Tailleur, 1987]. The central Brooks Range sites all have normal polarity overprints and appear to yield poles that lie close to the expected Cretaceous standstill position of 80-120 my [Globerman and Irving, 1988], which corresponds with the Cretaceous long normal polarity interval.

Paleozoic carbonate rocks in the northeastern Brooks Range in the vicinity of the Sadlerochit Mountains has also been magnetically overprinted [Plumley and Tailleur, 1987]; however, both normal and reversed polarity characteristic components were observed which predate folding. In stratigraphic coordinates these poles lie in the approximate location of 60-40 my on the APWP. This agrees very well with the mid-Tertiary age of deformation and uplift for this part of the range [Lieggi, 1987].

The completion of thrusting and emergence for the Porcupine Terrane carbonate sequences is likely Late Cretaceous. By applying a regional fold test and observing the variation of kappa (directional dispersion) with incremental tilt corrections similar to that shown in Figure 11, it is found that there has been very little dispersion of site means since acquiring the characteristic remanent magnetization. The similarity of mean directions between the Upper and Lower Ramparts (Figure 12) further suggest

TABLE 2. Site Mean Directions. Study: Upper Ramparts

Average Site Long: -141.6 Lat: 67.2
Age: Silurian/Devonian Locality: Salmon Trout/Porcupine River Intersection

Characteristic Component				Geographic Directions with site statistics				Geographic Poles		Stratigraphic Directions and Poles			
#	Site	N/n	Dt	G_Dec	G_Inc	a95	k	G_Long	G_Lat	S_Dec	S_Inc	S_Long	S_lat
1	86P001	4/4	H	49.8	75.8	6.5	205	295.8	69.3	132.5	60.1	255.9	24.0
2	86P002	6/6	T	14.4	79.2	6.4	110	282.1	84.3	155.8	73.4	233.8	37.6
3	86P003	3/3	H	35.1	81.6	3.9	993	264.3	76.9	180.7	67.4	217.9	27.4
4	86P004	5/5	H	32.5	79.9	10.9	50	278.4	78.0	225.3	73.0	189.4	40.1
5	86P005	6/6	H	355.7	83.3	7.3	84	212.5	80.3	150.1	61.2	241.8	21.8
6	86P006	4/4	H	14.8	81.0	4.5	412	256.0	82.7	174.0	67.7	222.7	27.9
7	86P007	4/4	H	56.2	84.1	11.9	61	249.8	71.2	141.0	49.0	252.2	11.5
8	86P008	4/4	H	47.8	59.4	2.8	1072	329.9	52.6	101.9	56.3	282.2	29.3
9	86P009	6/6	T	47.0	56.2	4.7	202	333.3	49.8	95.7	53.2	289.0	28.7
10	86P010	4/4	T	31.1	75.7	6.0	238	311.6	76.4	2.8	3.9	35.3	24.7
11	86P011	5/5	T	13.3	70.1	12.1	41	6.0	75.4	332.9	73.9	110.2	76.1
12	86P012	4/4	T	24.4	78.7	9.6	93	290.9	80.8	329.0	80.1	160.3	78.5
13	86P025	4/4	H	88.8	75.0	11.2	68	273.3	54.7	181.7	73.7	217.3	36.9
14	86P026	4/4	H	56.8	75.4	6.2	222	292.5	66.3	201.6	87.2	214.0	61.9
15	86P027	4/4	H	22.3	77.4	8.2	128	306.3	81.1	256.1	73.6	170.6	48.4
16	86P035	4/4	T	158.6	86.2	19.2	24	223.6	60.0	12.4	35.2	22.4	41.6
17	86P036	4/4	H	43.1	86.6	6.1	230	233.2	71.6	20.6	31.9	13.0	38.3
18	86P037	4/4	H	331.6	86.2	4.9	353	205.7	73.5	16.5	34.4	17.7	40.6
19	86P038	5/4	T	134.7	84.8	25.6	14	231.0	60.4	357.6	23.0	41.0	35.4
20	86P039	5/5	H	78.4	77.0	17.0	21	274.7	60.4	13.1	14.0	23.5	29.2

MEAN OF GEOGRAPHIC DIRECTIONS

N/n	Dec	Inc	A95	k	R	sd		
20/20	42.0	79.2	4.1	64.4	19.70492	10.1	Colat =	20.9

MEAN OF VGP'S FROM GEOGRAPHIC DIRECTIONS

N/n	Long	Lat	A95	K	R	sd		
20/20	277.7	75.0	6.9	23.2	19.17971	16.8	Colat =	19.7

MEAN OF STRATIGRAPHIC DIRECTIONS

N/n	Dec	Inc	A95	k	R	sd		
20/20	44.4	76.6	18.0	4.3	15.55300	39.2	Colat =	25.5

MEAN OF VGP'S FROM STRATIGRAPHIC DIRECTIONS

N/n	Long	Lat	A95	K	R	sd		
20/20	261.4	68.9	26.0	2.6	12.57285	50.6	Colat =	15.8

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TABLE 2. Continued

Secondary Component				Geographic Directions with site statistics				Geographic Poles		Stratigraphic Directions and Poles			
#	Site	N/n	Dt	G_Dec	G_Inc	a95	k	G_Long	G_Lat	S_Dec	S_Inc	S_Long	S_lat
1	86P001	4/4	H	0.0	72.7	7.6	148	38.4	80.9	139.3	73.3	244.3	39.7
2	86P002	6/6	T	24.0	80.9	7.9	73	266.3	80.4	155.7	71.0	234.7	34.0
3	86P003	3/3	H	195.6	85.6	5.9	436	213.9	58.7	190.8	55.6	209.4	13.7
4	86P004	5/5	H	39.4	78.2	4.2	470	289.7	75.0	224.5	72.9	189.8	39.8
5	86P005	6/6	H	338.0	81.0	10.0	60	172.9	80.9	153.9	64.0	238.2	24.6
6	86P006	4/4	H	17.4	81.7	8.3	124	252.1	81.3	173.7	66.8	223.0	26.7
7	86P007	4/4	H	229.1	85.3	20.7	21	204.1	60.3	156.9	47.9	238.7	7.7
8	86P008	4/4	H	54.5	65.5	12.0	60	316.1	56.4	114.1	56.1	272.3	25.0
9	86P009	6/6	H	45.7	64.5	5.4	154	326.8	58.6	108.2	57.4	276.5	28.2
10	86P010	4/4	T	32.2	74.4	10.7	75	317.6	74.7	3.9	3.2	34.1	24.3
11	86P011	5/5	T	359.3	77.9	11.8	43	72.6	89.5	300.4	75.8	147.5	65.7
12	86P012	4/4	T	23.1	83.4	7.6	149	243.8	78.1	301.3	81.6	172.7	70.2
13	86P025	4/4	H	64.9	78.2	28.3	12	277.8	66.1	202.8	77.2	205.6	43.7
14	86P026	4/4	H	95.8	79.8	19.7	23	258.8	58.7	194.5	78.0	210.5	44.6
15	86P027	4/4	H	318.5	81.5	16.2	33	170.8	75.1	251.3	62.1	163.3	32.9
16	86P035	4/4	T	4.6	80.9	11.3	67	233.5	84.7	9.1	22.9	27.3	34.4
17	86P036	4/4	H	4.7	84.3	3.5	675	223.0	78.4	17.4	29.5	17.2	37.4
18	86P037	4/4	H	328.7	87.1	12.8	53	208.7	71.9	17.2	35.1	16.7	40.9
19	86P038	6/6	T	299.2	80.0	11.7	34	165.0	68.8	345.7	13.1	54.6	28.7
20	86P039	5/4	H	349.7	85.4	14.7	40	211.6	76.1	359.2	12.4	39.3	29.1

MEAN OF GEOGRAPHIC DIRECTIONS

N/n	Dec	Inc	A95	k	R	sd		
20/20	22.3	82.2	3.9	72.2	19.73699	9.5	Colat =	15.4

MEAN OF VGP'S FROM GEOGRAPHIC DIRECTIONS

N/n	Long	Lat	A95	K	R	sd		
20/20	248.4	79.6	7.1	22.3	19.14660	17.2	Colat =	14.7

MEAN OF STRATIGRAPHIC DIRECTIONS

N/n	Dec	Inc	A95	k	R	sd		
20/20	28.2	81.2	19.8	3.7	14.84152	42.2	Colat =	17.2

MEAN OF VGP'S FROM STRATIGRAPHIC DIRECTIONS

N/n	Long	Lat	A95	K	R	sd		
20/20	234.5	68.3	28.2	2.3	11.79596	53.2	Colat =	6.2

N/n: total number of samples or sites/number of samples of sites included in analysis.
Dt: demagnetization type, (T) thermal or (H) alternating field.
a95: 95% confidence radius.
k, K: precision parameter of directions, poles.
G_Dec, G_Inc, G_Long, G_Lat: Geographic coordinates for directions (Declination, Inclination) and poles (Longitude, Latitude).
R: resultant vector length.
sd: standard deviation.
Colat: angle between study site and computed pole.
VGP: virtual geomagnetic pole.

that the remagnetization event post-dates the apparent block rotations which are documented in the structural data. Finally, the remagnetization event must pre-date the Porcupine River basalt field from which paleomagnetic directions yield an expected late Tertiary pole and thereby place a lower time constraint on tectonic rotations.

Site 86P038 is the only known Paleozoic carbonate strata in Northern Alaska that records a magnetization that predates the Brookian deformation. The third component which is present in all six core samples analyzed, yields site-mean directions (Geographic Coord: Decl=232, Incl=45; Stratigraphic Coord: Decl=306, Incl=36, a95=20) which are distinct from that expected for the Mesozoic-Cenozoic period. This study is insufficient in sample size to establish primacy of magnetization and to compute a paleomagnetic pole, however due to the significance of this

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result, strata of similar facies type will be pursued in the coming northern Alaska field seasons.

Discussion

The Upper and Lower Ramparts appear to be related stratigraphically and share many of the same structural relationships. The only structural evidence for a pre-Carboniferous deformational event is the unconformity between the Mississippian and Devonian strata seen by Laudon et al. [1966] and Brosge' et al. [1966]. Understanding the true nature of this unconformity is critical to our analysis in determining the number of major deformational events and to the correct age assignment of regional events experienced by the Porcupine Terrane. Whether or not this contact is a disconformity, and the result of epeirogenic movement, is unresolved at this time. Nevertheless, the Mississippian is multiply deformed, while the Permian is not. We therefore tentatively assign the age of D1, which documents north-south to northwest-southeast lateral compression, to Late Carboniferous.

In the Franklin Mountains of the northeastern Brooks Range, Oldow et al. [1987] have documented a southeastward shortening event in the mid-Devonian (Ellesmerian Orogeny) and a north-verging, Cretaceous-Tertiary event that is related to the Brookian orogeny. Bell [1974] has likewise found evidence for a pre-Carboniferous event to the northeast of the Porcupine Terrane in the Yukon province that may be related to that seen by Oldow et al. However, the strong Late Carboniferous D1 event recorded in structural elements of the Upper Ramparts distinguishes this tectonostratigraphic terrane from the neighboring Paleozoic terranes to the north.

The post-Permian event, D2, which is contributed to Brookian Orogeny, is directed to the east and southeast. These structural orientations roughly parallel the northeastern Brooks Range flexure (Figure 2). From the structural data alone, it appears that the Upper Ramparts has rotated about 45 degrees clockwise with respect to the Lower Ramparts since generation of Brookian fabric.

The Paleozoic carbonate sequence has been remagnetized following D2 deformation, and the age of acquisition of this overprint is probably early to mid Tertiary. Although the mean characteristic direction at the 95% confidence

TABLE 3. Site Mean Directions. Study: Lower Ramparts

Average Site Long: -142.6 Lat: 67.0
Age: Devonian Locality: Lower Ramparts, Porcupine River

Characteristic Component				Geographic Directions with site statistics				Geographic Poles		Stratigraphic Directions and Poles			
#	Site	N/n	Dt	G_Dec	G_Inc	a95	k	G_Long	G_Lat	S_Dec	S_Inc	S_Long	S_lat
1	86P042	4/4	T	274.3	83.1	9.8	89	184.6	64.4	329.2	82.5	181.0	77.3
2	86P043	4/4	T	319.4	84.8	4.2	481	193.2	73.5	317.3	74.8	131.0	71.1
3	86P044	13/12	T	291.0	85.6	3.5	151	194.5	68.6	311.8	75.0	136.2	69.1
4	86P045	4/4	H	29.4	87.2	10.2	83	226.1	71.7	345.8	68.1	68.5	72.7
5	86P046	4/4	H	351.4	81.6	11.4	66	197.5	82.9	343.5	61.7	66.1	64.3
6	86P047	5/5	H	266.5	89.5	6.2	155	214.9	66.9	338.7	69.9	85.7	73.3
7	86P048	5/5	H	22.3	66.7	8.0	92	353.6	69.0	355.2	47.9	44.0	51.8
8	86P049	4/4	H	21.5	65.3	5.7	260	357.1	67.5	75.6	79.5	267.0	63.8
9	86P050	5/4	T	35.7	68.1	10.5	78	331.3	66.4	49.5	63.9	321.9	56.6
10	86P051	7/6	T	86.3	64.2	9.6	49	287.9	42.8	356.6	75.6	57.6	85.6

MEAN OF GEOGRAPHIC DIRECTIONS

N/n	Dec	Inc	A95	k	R	sd		
10/10	26.9	80.9	8.1	36.3	9.75191	13.4	Colat =	17.8

MEAN OF VGP'S FROM GEOGRAPHIC DIRECTIONS

N/n	Long	Lat	A95	K	R	sd		
10/10	256.1	79.6	14.4	12.2	9.25979	23.2	Colat =	16.2

MEAN OF STRATIGRAPHIC DIRECTIONS

N/n	Dec	Inc	A95	k	R	sd		
10/10	353.1	72.7	9.0	30.0	9.69973	14.8	Colat =	31.9

MEAN OF VGP'S FROM STRATIGRAPHIC DIRECTIONS

N/n	Long	Lat	A95	K	R	sd		
10/10	66.2	82.1	14.3	12.4	9.27177	23.0	Colat =	30.1

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TABLE 3. Continued

Secondary Component				Geographic Directions with site statistics				Geographic Poles		Stratigraphic Directions and Poles			
#	Site	N/n	Dt	G_Dec	G_Inc	a95	k	G_Long	G_Lat	S_Dec	S_Inc	S_Long	S_lat
1	86P042	4/4	T	65.7	84.1	15.0	38	248.5	69.1	45.8	78.1	285.7	72.6
2	86P043	4/4	T	329.8	85.8	6.9	180	202.3	73.7	320.3	75.9	133.6	73.2
3	86P044	13/13	T	331.7	83.3	9.7	19	188.3	77.1	324.5	72.3	114.1	71.3
4	86P045	4/4	H	82.8	89.3	6.5	198	221.0	67.1	342.0	70.1	80.0	74.5
5	86P046	4/4	H	39.6	81.1	10.1	84	267.2	75.6	358.0	64.3	41.3	69.1
6	86P047	5/3	H	205.9	79.1	27.9	21	204.1	47.1	307.3	75.4	141.0	67.8
7	86P048	5/5	H	346.2	72.1	12.0	42	76.7	78.3	334.7	47.9	72.0	49.0
8	86P049	4/4	H	21.8	76.6	12.1	59	312.7	80.8	141.9	79.6	236.4	49.3
9	86P050	5/5	T	11.4	86.1	37.0	5	222.9	74.5	68.6	81.8	259.6	67.5
10	86P051	7/7	T	58.1	82.6	8.6	49	256.8	70.5	310.1	63.3	112.0	55.8

MEAN OF GEOGRAPHIC DIRECTIONS

N/n	Dec	Inc	A95	k	R	sd		
10/10	11.6	85.2	5.0	92.9	9.90311	8.4	Colat =	9.5

MEAN OF VGP'S FROM GEOGRAPHIC DIRECTIONS

N/n	Long	Lat	A95	K	R	sd		
10/10	225.4	75.9	9.6	26.2	9.65608	15.8	Colat =	9.2

MEAN OF STRATIGRAPHIC DIRECTIONS

N/n	Dec	Inc	A95	k	R	sd		
10/10	338.8	75.2	9.9	24.9	9.63833	16.2	Colat =	27.8

MEAN OF VGP'S FROM STRATIGRAPHIC DIRECTIONS

N/n	Long	Lat	A95	K	R	sd		
10/10	119.5	81.4	16.6	9.5	9.04782	26.3	Colat =	25.5

Symbols same as Table 2.

level of the Upper Ramparts is not statistically distinct from the mean characteristic direction of the Lower Ramparts, the steep inclination of the magnetic overprint nearly precludes the use of paleomagnetic directional data to discriminate structural rotations between the two regions.

Hence, characteristic magnetization from all sites of the Upper and Lower Ramparts have been combined to produce a mean direction and mean pole. This pole does not match any reference position on the North American APWP. This differs from many of the previous paleomagnetic studies of the Brooks Range strata [Hillhouse and Gromme, 1983; Plumley and Tailleur, 1987; Plumley and Reusing, 1984] which give remagnetization poles near the Cretaceous paleomagnetic reference pole for North America.

The mean paleomagnetic pole from the carbonate sequence lies off in a region that suggests clockwise displacement of the Porcupine Terrane with respect to North America. Simple tilt about a horizontal axis could also produce this pole position, but the sampled region spans too great an area for such a mechanism to apply. Furthermore, the tilt expected from the mapped normal faults predicts a displacement of the pole into the region of the Pacific Ocean, which is the opposite of what is observed.

The amount of rotation of the Porcupine carbonate strata, if about a vertical axis, will vary depending on the age of remagnetization. The older the age, the greater the amount of rotation. If the characteristic component is as young as 30 my, then the rotation is approximately 45 degrees. However, if it is as old as early Tertiary or Late Cretaceous, a rotation of 90 degrees or more is indicated by this study (Figure 13).

Although the interpreted age of the overprint is rather tenuous, the age constraints suggest clockwise rotations of 45 to 90 degrees. However, a reverse rotation of this magnitude would essentially straighten the northeast Brooks Range flexure. The D2 deformational event documented in the Ramparts region would then parallel the trend of Late Cretaceous structural elements that are observed in the central Brooks Range. The D1 deformational event would likewise be directed east-west and be orthogonal to the Paleozoic coast line of northwest North America.

Two large sedimentary basins lie south of the Brooks Range; the Yukon-Koyukuk basin and the Yukon Flats Basin (Figure 2). Coney and Jones [1985] suggest that both basins are underlain by early Mesozoic, mafic oceanic crust. Numerous normal faults are observed in the Porcupine Terrane that cut all previously mentioned structure and the majority of these dip toward the Yukon Flats region. We suggest that the Yukon Flats Basin, unlike the Yukon-Koyukuk, is a

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TABLE 4. Mean Direction and Paleomagnetic Pole for Porcupine River Carbonate Section.

Locality: Upper and Lower Ramparts of the Porcupine River

Characteristic Component

MEAN OF GEOGRAPHIC DIRECTIONS								
N/n	Dec	Inc	A95	k	R	sd		
30/30	37.6	79.8	3.7	52.5	29.44731	11.2	Colat =	19.7
MEAN OF VGP'S FROM GEOGRAPHIC DIRECTIONS								
N/n	Long	Lat	A95	K	R	sd		
30/30	272.3	76.6	6.4	18.1	28.39894	19.0	Colat =	18.7
MEAN OF STRATIGRAPHIC DIRECTIONS								
N/n	Dec	Inc	A95	k	R	sd		
30/30	21.8	76.5	11.9	5.9	25.08385	33.4	Colat =	25.7
MEAN OF VGP'S FROM STRATIGRAPHIC DIRECTIONS								
N/n	Long	Lat	A95	K	R	sd		
30/30	267.1	81.0	17.3	3.3	21.17599	44.7	Colat =	18.5

Secondary Component

MEAN OF GEOGRAPHIC DIRECTIONS								
N/n	Dec	Inc	A95	k	R	sd		
30/30	19.8	83.2	3.0	78.3	29.62949	9.2	Colat =	13.4
MEAN OF VGP'S FROM GEOGRAPHIC DIRECTIONS								
N/n	Long	Lat	A95	K	R	sd		
30/30	239.1	78.6	5.5	23.5	28.76658	16.7	Colat =	13.1
MEAN OF STRATIGRAPHIC DIRECTIONS								
N/n	Dec	Inc	A95	k	R	sd		
30/30	2.4	79.8	12.9	5.1	24.36798	35.7	Colat =	19.7
MEAN OF VGP'S FROM STRATIGRAPHIC DIRECTIONS								
N/n	Long	Lat	A95	K	R	sd		
30/30	216.6	78.7	18.6	3.0	20.30256	46.8	Colat =	11.7

Symbols same as Table 2.

recent extensional basin forming in response to post-Brookian oroclinal bending.

Summary

We have been unsuccessful at recovering a primary magnetic remanence in a sufficient number of Paleozoic samples from which to compute a paleolatitude. Therefore, the position of the Porcupine Terrane with respect to North America during the Paleozoic is still unknown. Similarity of stratigraphic sections and timing of deformational events favor association of the Porcupine Terrane strata with the shelf carbonates of the northern Canadian Cordillera.

In light of the orientation of structural fabric and directions of paleomagnetic overprints presented in the previous sections, the succession of deformational events which have affected the Porcupine River area can be summarized as follows. Late Carboniferous deformation (D1) developed north-south trending, east-verging, thrusts and folds throughout the Porcupine Terrane. The Cretaceous Brookian Orogeny produced northeast-verging thrusts and folds (D2), accompanied by the emplacement of allochthonous thrust sheets of oceanic affinity onto the Paleozoic section. The Porcupine megashear developed after the Brookian Orogeny as a consequence of the convergence ot the North American-Eurasian crustal plates [Harbert et al., 1987]. As a result, the northeast Brooks Range was bisected producing relative rotations between crustal blocks in the vicinity of the Porcupine megashear zone. Complete remagnetization of the Paleozoic carbonate strata also followed the Brookian deformational event and may even post-date the major strike-slip activity on the megashear.

Regional compressional tectonics resulted in the formation of the Cretaceous Brookian fold and thrust belt. We propose that the structural grain of the former Brooks Range was roughly aligned with the present central Brooks Range and northern Canadian Cordilleran trends. Subsequent bending in the Tertiary produced the eastern Brooks Range orocline. This flexural bending may be partly caused by the Porcupine megashear.

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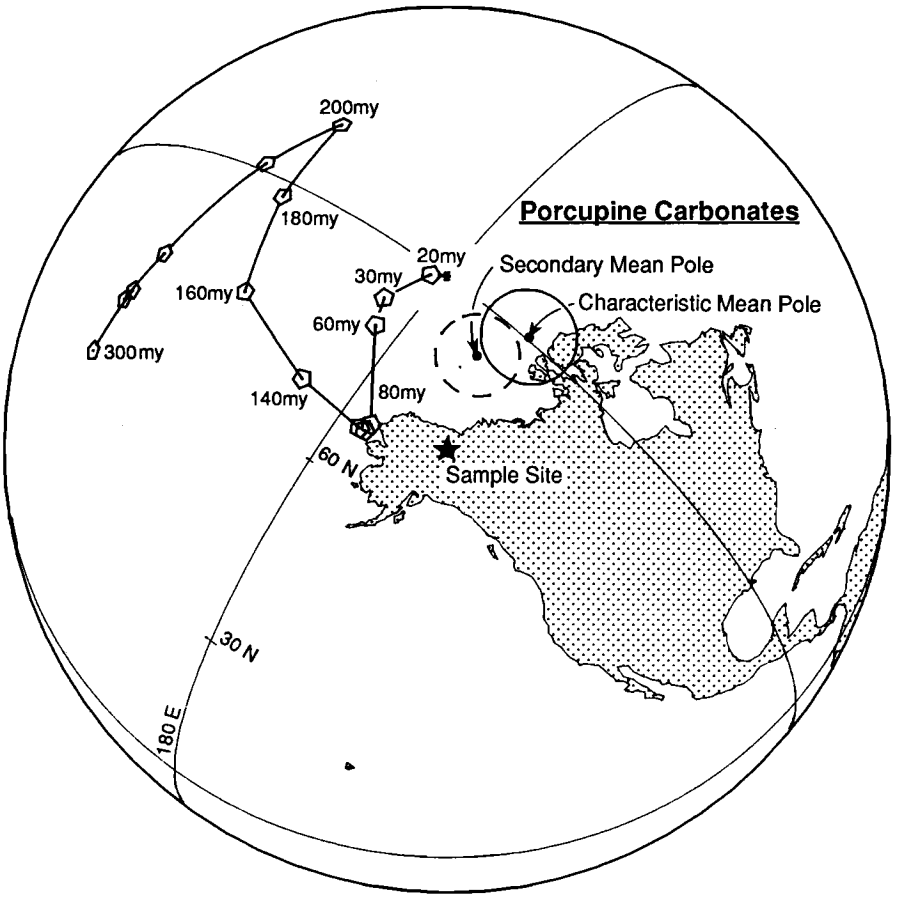


Fig. 13. Plot of the characteristic and secondary mean poles for the Porcupine carbonates (uncorrected geographic coordinates) and the Apparent Polar Wander Path (APWP) for North America. Circle of confidence for the pole of the characteristic magnetization is a solid circle and dashed circle for the secondary pole. Ages of the APWP pentagons are labeled. The Tertiary reference APWP is constructed as follows: the 20 my pole is from Irving and Irving [1982] and is an average of a 30 my window. The 30 my pole is from Diehl et al. [1988] and is a 20 my window. The 60 my reference pole (163.3E, 78.8 N, a95=5.0, N=9) is an average of Irvings 60 my reference pole (20 my window) and recent Paleocene data of Diehl et al. [1983], and Gordon [1984]. The 80 my, 100 my, and 120 my, reference poles are from Irving and Irving [1982].

However, important regional kinematic constraints that must be satisfied in any model explaining the generation of the northeast Brooks Range flexure are 1) early to middle Tertiary clock-wise rotation of the southeastern limb of the flexure, 2) Tertiary extension in the Yukon Flats region with accompanying flood basalts, and 3) continued northward migration of an active deformational front in the northeasternmost reaches of the Brooks Range.

A model of northward arc migration in a continental setting suggests an analogy to A-Type subduction as proposed by Bally [1975]. This mechanism has been applied to explain numerous tight oroclinal features in the western Mediterranean [Malinverno and Ryan, 1986; Burchfiel, 1980]. The Yukon Flats Basin becomes, by inference, an intracratonic, back-arc basin.

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